Chapter 1

Introduction to Communication Systems

1. History of Communication

A time chart showing the historical development of communications is given in Table 1.1. The reader is encouraged to spend some time studying this table to obtain an appreciation for the chronology of communications.

Table 1.1 History of Communication

Year	Event
1834	Carl F. Gauss and Ernst H. Weber build the electromagnetic telegraph
1838	William F. Cooke and Sir Charles Wheatstone build the telegraph
1844	Samuel F. B. Morse demonstrates the Baltimore, MD, and Washington, DC, telegraph line
1858	The first transatlantic cable is laid, and fails after 26 days
1864	James C. Maxwell predicts electromagnetic radiation
1871	The society of telegraph Engineers is organized in London
1876	Alexander Graham Bell develops and patents the telephone
1887	Heinrich Hertz verifies Maxwell's theory
1889	The Institute of Electrical Engineers (IEE) forms from the society of Telegraph Engineers in London
1894	Oliver Lodge demonstrates wireless communication over a distance of 150 yards
1900	Guglielmo Marconi transmits the first transatlantic wireless signal
1905	Reginald Fessenden transmits speech and music by radio
1915	Bell system completes a U.S. transcontinental telephone line
1918	Edwin H. Armstrong invents the super-heterodyne receiver circuit
1920	KDKA, Pittsburgh, PA, begins the first scheduled radio broadcasts
1920	J.R. Carson applies sampling to communications
1926	J.L. Baird, England and C.F. Jenkins, United States, demonstrate television
1931	Teletypewriter service is initiated
1933	Edwin H. Armstrong invents FM
1935	Robert A. Watson-Watt develops the first practical radar
1936	The British Broadcasting corporation (BBC) begins the first television broadcasts
1937	Alex Reeves conceives pulse code modulation (PCM)
1941	John V. Atanasoff invents the computer at Iowa State College.
1947	Steve O. Rice develops statistical representation for noise at bell laboratories
1948	Claude E. Shannon publishes his work on information theory
1950	Time-division multiplexing is applied to Telephony
1950	Microwave telephones and communication links are developed
1953	NTSC color television is introduced in the United States
1953	The first transatlantic telephone cable (36 voice channels) is laid
1957	The first Earth satellite, Sputnik I, is launched by USSR
1961	Stereo FM broadcast begin in the United States
1962	The first active satellite, Telstar I, relays television signals between the U.S. and Europe
1963	Bell system introduces the touch-tone phone
1964	The electronic telephone switching system is placed into service
1965	The first commercial communication satellite, Early Bird, is placed into service
1968	Cable television systems are developed

1972	Motorola demonstrates the cellular telephone to the FCC
1980	Bell system FT3 fiber optic communication is developed
1985	Fax machines become popular
1989	Pocket cellular telephone is introduced by Motorola
1990---	Era of digital signal processing with microprocessors, digital oscilloscopes, digitally tuned receivers, ISDN, digital satellite systems and high definition television (HDTV), etc.

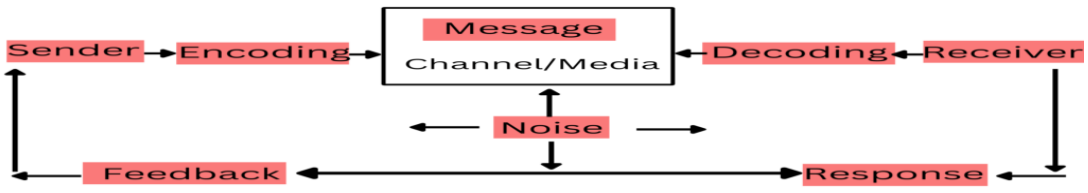
2. Communication Process

The term communication covers very broad fields of study ranging from the use of symbols to large social implications and effects. Shortly, communication means the transmission of information from one point to another. At one time, it would have been called telecommunications, using the Greek prefix tele-, meaning far. Applications of communication enable us to communicate with people all over the globe by the help of mobile phones, radios, televisions, and computers connected to internet in our offices. Thus, communication affects our daily life in so many different ways that it is easy to overlook several of its aspects. Communication provides information to ships on seas, aircraft in flight, rockets and satellites in space. It also keeps a weather forecaster informed of environmental conditions measured by a multitude of sensors. Indeed, a list of applications involving the use of communication in one way or another is almost endless.

3. Elements of communication system

Seven major elements of communication process are: (1) sender (2) ideas (3) encoding (4) communication channel (5) receiver (6) decoding and (7) feedback.

Communication may be defined as a process concerning exchange of facts or ideas between persons holding different positions in an organization to achieve mutual harmony. The communication process is dynamic in nature rather than a static phenomenon. Communication process as such must be considered a continuous and dynamic inter-action, both affecting and being affected by many variables.



Block diagram of communion system

(1) Sender:

The person who intends to convey the message with the intention of passing information and ideas to others is known as sender or communicator.

(2) Ideas:

This is the subject matter of the communication. This may be an opinion, attitude, feelings, views, orders, or suggestions.

(3) Encoding:

Since the subject matter of communication is theoretical and intangible, its further passing requires use of certain symbols such as words, actions or pictures etc. Conversion of subject matter into these symbols is the process of encoding.

(4) Communication Channel:

The person who is interested in communicating has to choose the channel for sending the required information, ideas etc. This information is transmitted to the receiver through certain channels which may be either formal or informal.

(5) Receiver:

Receiver is the person who receives the message or for whom the message is meant for. It is the receiver who tries to understand the message in the best possible manner in achieving the desired objectives.

(6) Decoding:

The person who receives the message or symbol from the communicator tries to convert the same in such a way so that he may extract its meaning to his complete understanding.

(7) Feedback:

Feedback is the process of ensuring that the receiver has received the message and understood in the same sense as sender meant it.

1.2 Sources of Information

Classification of Signals:

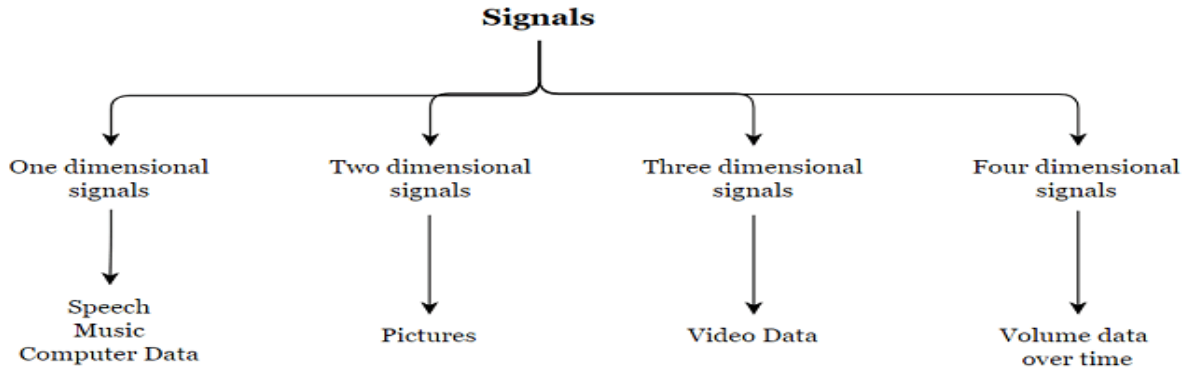
Information source in communication system

1. Speech:
2. Music:
3. Pictures:
4. Computer Data:

4. Signal and Classification of Signals

A signal may be defined as the single-valued function of time. Time plays the role of an independent variable. This means that at every instant of time, the signal has a unique value.

Classification of Signals:



5. Information source in communication system

Some of the important sources of information in the communication environment may be listed as under:

1. Speech	2. Music	3. Pictures	4. Computer data

A source of information is basically a signal which carries the information.

1. Speech:

Speech involves the transfer of information from the speaker to the listener. Such a transfer of information takes place in the following three stages:

1. Production
2. Propagation and
3. Perception

The band of frequencies considered to be essential for speech communication is 300 Hz to 3100 Hz. This band is utilized for commercial telephonic communication.

2. Music:

The music signal is originated from the instruments such as the piano, violin, flutes, etc. The musical note may last for a short time or for a longer time depending upon the instruments being used for its origination.

3. Pictures:

The picture can be either static or dynamic. An example of a static picture is a picture sent by a fax machine and that of a dynamic picture is one produced on T.V. The T.V. camera produces a dynamic picture and the picture tube reproduces it at the receiving end. The electrical equivalent of a dynamic picture is produced at the output of the camera and it is called the typical video signal. The **bandwidth** required for this video signal is **0 to 5 MHz**.

The video signal consists of a luminance signal (Y) and a chrominance signal. The luminance signal conveys the brightness information while the chrominance signal conveys the color information. Along with them, synchronization signals (horizontal and vertical) are also transmitted. These are used by the receiver to synchronize its vertical and horizontal scanning rates with those at the transmitter.

4. Computer Data:

Personal computers are used for electronic mail, the exchange of software, and sharing of resources. The text matter transmitted by a computer is encoded using the American standard code for Information Interchange (ASCII). Each character in ASCII is represented by seven data bits. Hence total $2^7 = 128$ characters can be represented using ASCII.

1.3 Shannon's Information capacity theorem

The Shannon capacity theorem defines the maximum amount of information, or data capacity, which can be sent over any channel or medium (wireless, coax, twister pair, fiber etc.).

$$C=B\log_2(1+S/N)$$

Where

C is the channel capacity in bits per second (or maximum rate of data)

B is the bandwidth in Hz available for data transmission

S is the received signal power

N is the total channel noise power across bandwidth B

What is Shannon's theorems in information theory?

In information theory, the Shannon–Hartley theorem tells the maximum entropy rate at which information can be transmitted over a communications channel of a specified bandwidth in the presence of noise.

Shannon Channel Capacity Theorem

The Shannon channel capacity theorem defines the theoretical maximum bit rate (number of bits per second) for a noisy channel

Capacity $C = B\ x\ log_2(1 + \frac{S}{N})$

C = Channel capacity (bps)
B = Bandwidth of signal
S = Signal power (Watt)
N = Noise power (Watt)

$$SNR = \frac{S}{N} = \frac{Signal\ Power}{Noise\ Power}\ (no\ unit)$$

Generally, SNR is expressed in decibel (dB) units. [dB = 10 $\log_{10}$ (S/N)]

Q. Calculate the bit rate for a noisy channel with SNR 300 and bandwidth of 3000Hz

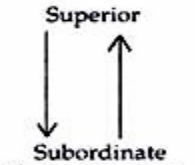
Solution: The bit rate for a noisy channel according to Shannon theorem can be calculated as follows:

$$Capacity\ C = B\ log_2\left(1 + \frac{S}{N}\right)$$
$$= 3000 * log_2(1 + 300)$$
$$= 3000\ log_2(301)$$
$$= 3000\ \frac{log_{10}(301)}{log_{10}(2)}$$
$$= 3000\ x\ \frac{2.4785}{0.3010}$$
$$= 3000 \text{ x } 8.23$$
$$= 24{,}690 \text{ bps}$$

The most frequently followed networks are the following:

1. Vertical Network:

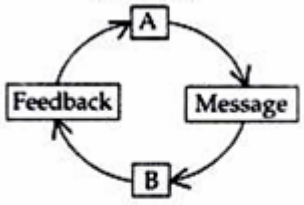
The vertical network is usually between the superior and subordinate and vice versa. It is two-way communication. The immediate feedback is possible in this type of communication network.



Vertical Network
Fig. 15.2

2. Circuit Network:

Under this network two persons communicate with each other. Say Mr. 'A' sends message to Mr. 'B'. After receiving message Mr. 'B' communicates the feedback message to Mr. 'A'. So communication takes the form of a circuit. Therefore it is known as circuit network. It is similar to vertical network but in circuit network 'A' and 'B' are not necessarily superior and subordinates.



(Circuit Network)
Fig 15.3

3. Chain Network:

This network of communication follows the organizational hierarchy and chain of command. All subordinates receive commands or instructions from their superior. B, C, D and E, F, G are the subordinates to A in the organizational hierarchy and receive commands from 'A' which follows the way shown in the diagram.

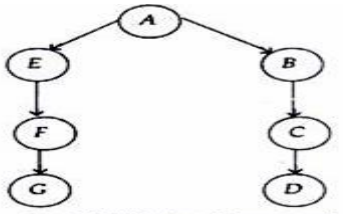


Fig. 15.4 Chain of Command

4. Wheel Network:

Here all subordinates receive commands from one superior. This is highly centralized type of communication network where each subordinate receives commands or instructions from a single authority or superior 'A' and wants the immediate feedback.

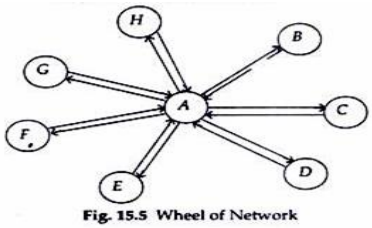


Fig. 15.5 Wheel of Network

5. Star Network:

Under star communication network all members of the group communicate with each other and exchange information. This network is a must for group communication or where teamwork is involved. This network channel of communication is open to all members of the group. The members communicate with each other without hesitation.

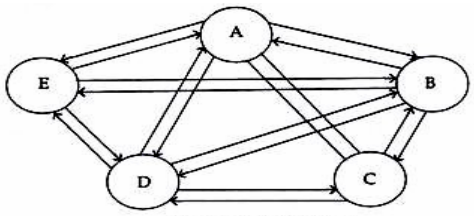
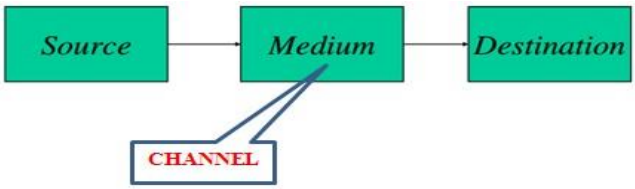


Fig. 15.6 Star Channel

The effectiveness of the above networks of communication channels depend upon their users i.e. the managers at all levels, their subordinates and other members of the organization and above all the seriousness with which all these human resources make use of the facilities provided to them by the organization to accomplish its objectives.

1.5 Communication Channels

The Channel provides the electrical connection between the source and destination. It is a physical path between transmitter and receiver.



A transmission medium (plural transmission media) is a material (solid, liquid or gas) which can propagate energy waves. For example, the transmission medium for sound received by the ears is air.

The quality of transmission is determined by both the characteristics of the medium and the characteristics of the signal. Transmission medium is a:

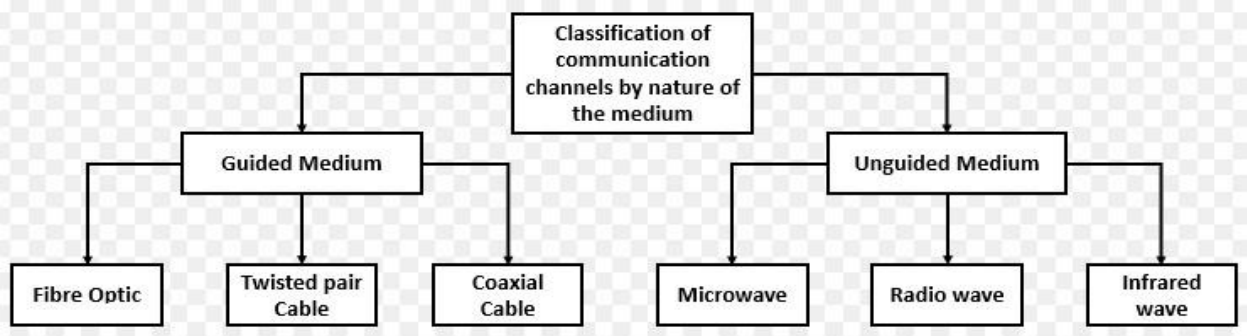
- ✓ Path between two telephones
- ✓ Path between two computers

- ✓ Path between the satellite and the ground station in satellite communication systems

There are two physical ways to transmit data: guided and unguided.

Guided media consists of physical connection between source and destination via a wire or a cable. In guided media, the signals are confined within the wire and guided along a physical path. Examples are twisted pair, coaxial cable and optical fiber.

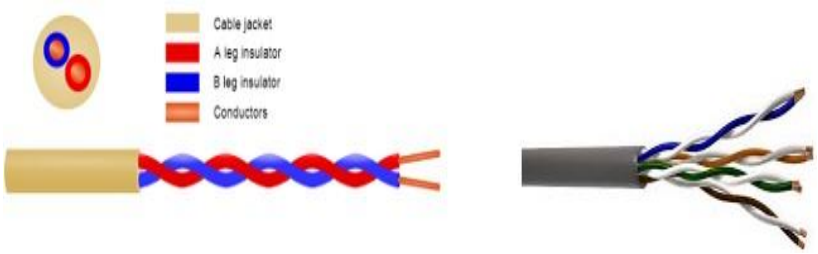
Unguided media transport electromagnetic (EM) waves without using a physical conductor. This type of communication is often referred to as wireless communication. Examples are atmosphere and outer space (Radio waves, microwaves, infrared) and satellite channels. Unguided (wireless) transmission is used in geographically disadvantaged places like: desert, forest, hills, mountains, oceans etc., where installing wires, cables is difficult.



Twisted Pair cable

Wire pairs are commonly used in local telephone communication and for short distance digital data communication. They are usually made up of copper and the pair of wires is twisted together. Data transmission speed is normally 9600 bits per second in a distance of 100 meter.

Commonly used medium and it is quite cheaper than any other transmission media. A twisted pair consists of two insulated conductors twisted together as shown in figure. Twisting is used to decrease interference from other wires. It can be shielded (cover for extra protection) or unshielded. The unshielded twisted pairs are very cheap and easy to install, but badly affected by noise and interference.



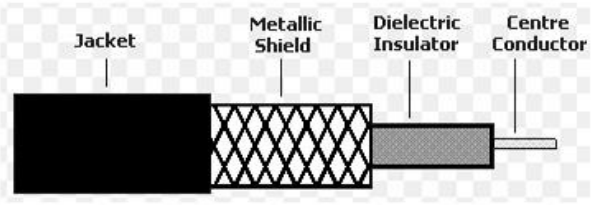
Coaxial cable

Coaxial cable consists of a central copper wire surrounded by an insulation over which copper mesh is placed. Coaxial cable is like a pipe with wire suspended in the middle of it. Due to its construction, coaxial cable is less susceptible to interference than twisted pair. Coaxial cable is capable of supporting a much higher frequency range than twisted pair.

One of the most popular use of co-axial cable is in cable TV (CATV) for the distribution of TV programmers. Another importance use of co-axial cable is in LAN and for long distance telephone communication.

Coaxial cable Features

- Two types of cables having 75 ohm and 50 ohm impedance are available
- Because of shielding, this cable has excellent noise immunity
- It has a large bandwidth and low losses
- Most widely used medium for Local Area Network (LANs)
- These cables are costlier than twisted pair cables, but cheaper than optical fiber cables





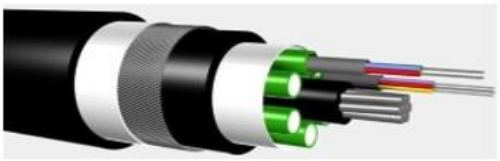
Fiber Optic cable

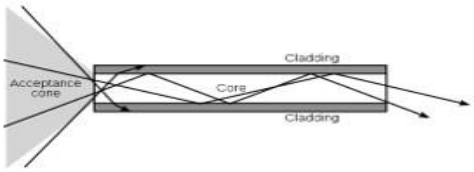
An optical fiber is constructed from a very thin stand of glass or ultra-pure plastic that is placed in a cladding of glass or plastic. Information is transmitted in the form of light, therefore no electrical interference and supports long distance communication. Note that twisted-pair cable, coaxial cable transport signals in the form of electric signals. Fibre optic operates at optical

frequencies (1014 to 1015 Hz). A fiber optic cable consists of a bundle of glass threads. The main disadvantage of fiber optics is that the cables are expensive to install.

Because of greater bandwidth (2Gbps), smaller diameter, lighter weight, low attenuation, immunity to electromagnetic interference (EMI) and longer repeater spacing, optical fiber cables are finding widespread use in long-distance telecommunications.

- Fiber optic carries information in the form of light
- Fiber optic cables have a much greater bandwidth than metal cables. This means that they can carry more data.
- Fiber optic cables are less susceptible to interference than other transmission medium.
- Fiber optic cables are much thinner and lighter than wires





Type of the channel	Frequency range (Approx.)
Twisted pair	1MHz – 100 MHz)
Coaxial cable	0 – 750 MHz
Microwave	1 GHz-30 GHz
Satellite	1 GHz – 40 GHz
Fibre optics	180 THz – 330 THz

1) **Microwaves-** In this kind the data is transferred via air. The waves travel in a straight line. The data is received and transferred via microwave stations. The speed at which data is transferred is 150 Mbps. They are widely used by telephone and cable companies.

2) **Satellite-** The signals are received from earth stations. Devices like GPS and PDAs also receive signals from these earth based stations. These satellites are located at a distance of 22300 miles above the earth. The process of transferring and receiving data takes place within few seconds. The data is transferred at a speed of 1 Gbps. They are used for purposes like weather forecast, military communication, radio transmission, satellite TV, data transmission, etc.

3) **Cellular Radio-** they are used for communication via mobile. High frequency radio waves are used for the transmission of data. You can receive and make calls and also access the internet.

4) **Radio Broadcasting-** Data is transferred and received via radio signals in the air. The transmission takes place for a long distance across cities or countries. The data is received and transferred via a transmitter. The speed at which data travels is 54 Mbps.